

DATA 201

Data Communication and Ethics

Aye Hninn Khine

What is research ethics?

Research ethics refers to principles and standards that guide responsible research conduct. In data science, this includes:

- honesty in reporting
- fairness in data use and analysis
- respect for data subjects and creators
- transparency in methods
- accountability for harm

Why ethics matters in data science

Data science can affect:

- privacy
- reputation
- employment
- access to services
- safety
- public trust in science

A technically strong project can still be ethically weak.

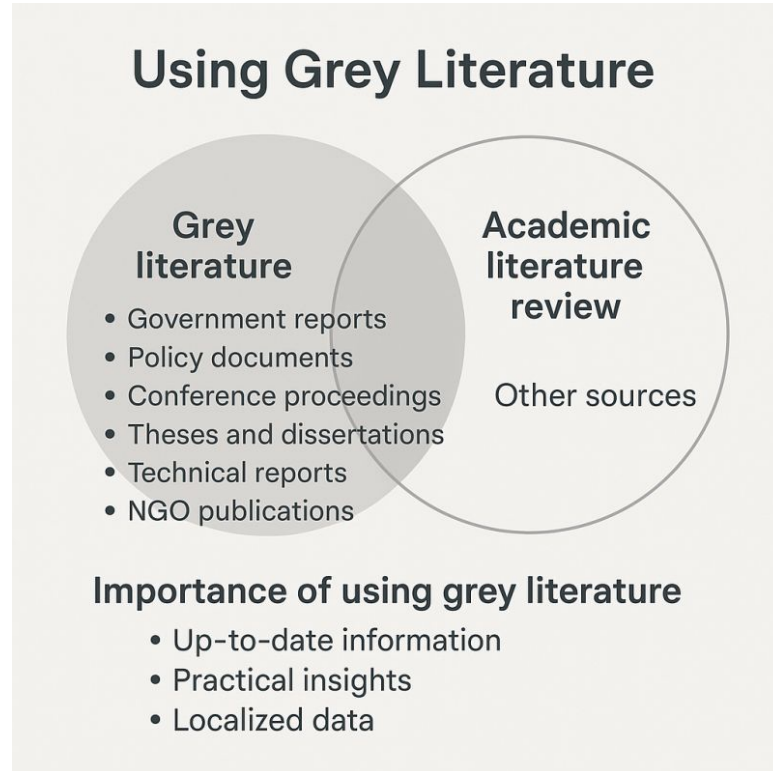
Literature Reviewing

Ethical literature reviewing means:

- citing original sources, not only secondary summaries
- representing prior work accurately
- not exaggerating weaknesses of others
- not ignoring relevant studies that contradict your idea
- not claiming novelty when similar work already exists

Recommended Practices for Literature Review

- Citing peer-reviewed articles
- Avoid unconfirmed sources



Common problems

citation padding: adding references not actually used

citation copying: citing papers only because another paper cited them

selective citation: only citing work that supports your claim

false novelty: pretending the work is new when it is not

Extending the work with proper credit

What counts as extending prior work?

Possible extensions:

- applying a method to a new dataset
- evaluating on a new language or domain
- improving efficiency
- improving interpretability
- addressing fairness or robustness
- reproducing and validating claims

Students must clearly acknowledge:

- the original idea
- the original dataset
- the original codebase or framework
- borrowed figures, tables, prompts, or evaluation setup
- prior benchmarks they build upon

Extending the work with proper credit

3. What is not acceptable?

- changing the name of an existing dataset and presenting it as new
- running new benchmarks on an old dataset without clearly crediting the creators
- reusing code and removing attribution
- slightly rewording others' ideas and claiming originality
- presenting replication as innovation

Copyright (Understanding License)

For code

- **MIT**: permissive, usually allows reuse with attribution
- **Apache 2.0**: permissive, includes patent terms
- **GPL**: strong copyleft, derivative works may need same license
- **BSD**: permissive

For creative content/data

- **CC BY**: use allowed with attribution
- **CC BY-SA**: attribution + share alike
- **CC BY-NC**: non-commercial only
- **CC BY-ND**: no derivatives
- **CC0**: public domain style release
- **All rights reserved**: no permission unless explicitly granted

Copyright (Understanding License)

You should always ask:

- Who created this?
- What is the license?
- Can I reuse it?
- Can I modify it?
- Can I redistribute it?
- Can I use it for commercial purposes?
- Do I need attribution?
- Are there ethical restrictions even if the license allows use?

Data Collection Ethics

Plagiarism Using others' text, ideas, figures, or results without proper credit.

Self-plagiarism Reusing your own previous writing or results without transparency.

Fabrication Making up data.

Falsification Manipulating data, methods, or results.

Cherry-picking Showing only the best-performing model or subset.

Salami slicing Splitting one meaningful study into multiple weak papers to inflate publication count.

Duplicate submission Submitting the same paper to multiple venues at the same time where prohibited.

Publication Ethics

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Authorship Ethics

A legitimate author usually contributes substantially to:

- study design
- data collection or curation
- analysis or interpretation
- writing or major revision
- approval of the final version
- responsibility for the work

Unethical authorship practices

- gift authorship: adding a senior person who did not contribute
- ghost authorship: excluding someone who did major work
- coercive authorship: adding a person because of power pressure
- honorary authorship: adding someone for prestige

Acknowledgment vs authorship

Someone should usually be acknowledged, not listed as author, if they only:

- provided funding
- gave general advice
- proofread language
- allowed use of lab space
- gave administrative support

References

- Kumar, R. (2019). *Research Methodology: A Step-by-Step Guide for Beginners* (5th ed.). SAGE Publications Ltd.

Thank You

ayehninnkhine@parami.edu.mm